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COURSE: Autonomous Systems Lab 1

GROUP: 4

TRUCK PLATOONING SCENARIO: Emergency Braking

Scenario Context

A truck platoon consisting of one lead truck and two follower trucks is travelling at normal cruising speed on a highway. The lead truck's onboard sensors detect an obstacle on the road that requires an immediate full stop. The lead truck applies maximum braking and simultaneously broadcasts an emergency brake signal to all following trucks via V2V (Vehicle-to-Vehicle) communication. Each follower truck receives the signal and applies emergency braking without waiting for a human reaction, allowing the entire platoon to stop in a coordinated cascade. The Platoon Manager monitors the braking status of all trucks and confirms the full stop before broadcasting a standby command.

If V2V communication is unavailable, follower trucks fall back to onboard sensor-based collision avoidance to detect the deceleration of the truck ahead and apply braking independently.

The platoon must:

1. Detect the obstacle via onboard sensors
2. Broadcast the emergency brake signal via V2V
3. Apply coordinated braking across all trucks
4. Confirm full stop and enter standby mode
5. Fall back to sensor-based braking if V2V fails

Timing Phases

PHASE	DESCRIPTION	TYPICAL TIMING CHARACTERISTICS
1	Hazard detection by lead truck sensors	<= 50 ms
2	Emergency brake signal broadcast via V2V	<= 100 ms
3	Follower truck brake activation	<= 150 ms from signal
4	Sensor-based fallback activation	<= 200 ms (if V2V unavailable)
5	Full platoon deceleration	1 – 5 seconds
6	Complete stop and standby confirmation	<= 5 seconds from signal

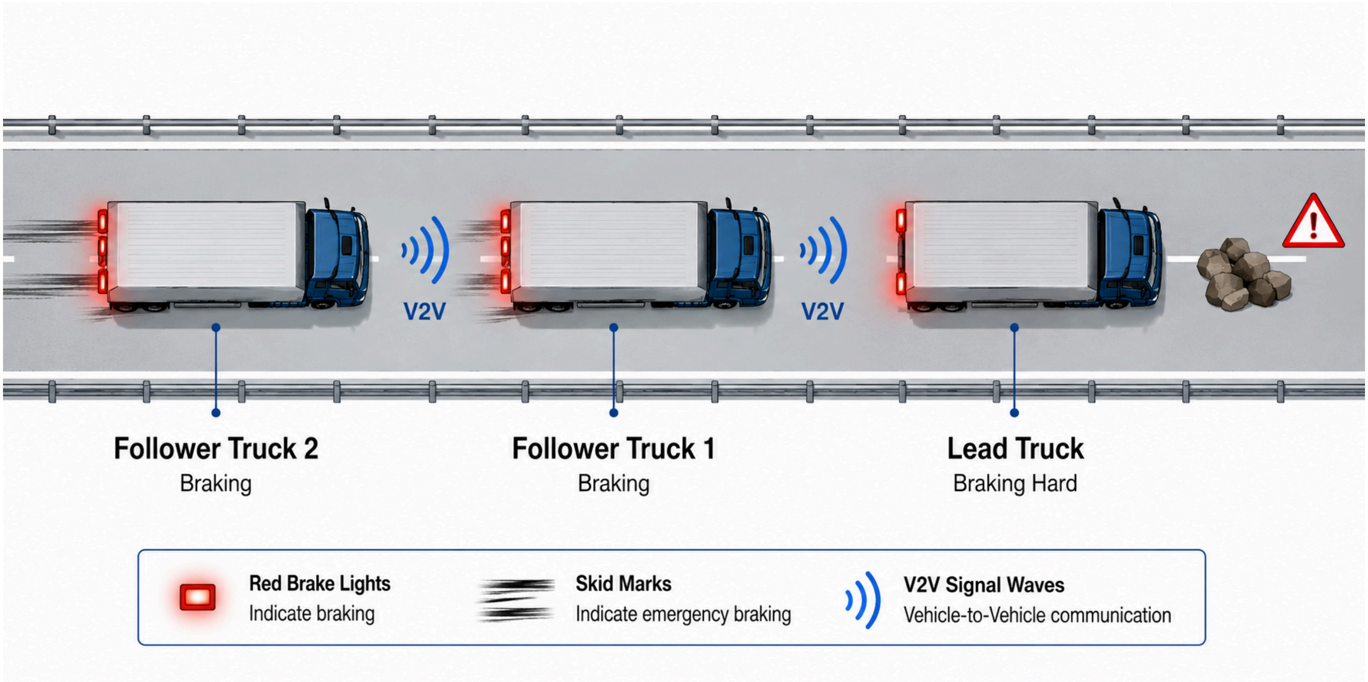

Fig.1 Emergency Braking Scenario Illustration (this image was generated by ChatGPT)